HACKER vs DEFENDER

“Can your defense move faster than the attacker can adapt?”

A real-time, AI-powered cybersecurity battle simulator. A Hacker agent and a Defender agent compete head-to-head while human users step in as Security Manager — making the calls the algorithms can’t make alone.



HACKER
attacks

DEFENDER
responds

SECURITY MANAGER
decides

SDGs & Targets

Where this project contributes to global goals

9



Industry, Innovation & Infrastructure

Strengthens resilient digital infrastructure by giving organizations a proactive way to test and harden cyber-defenses.

16



Peace, Justice & Strong Institutions

Supports safer digital institutions by reducing cybercrime impact through faster, smarter incident response.

4



Quality Education

Builds hands-on, gamified cybersecurity skills for students and security teams through simulated attack scenarios.

Problem Statement

Why traditional cyber-defense training falls short

01

Reactive, not proactive

Most tools raise an alert only after suspicious activity has begun — defenders are always one step behind.

02

Attackers adapt fast

Modern attacks probe, learn, and change tactics in real time, while human response cycles stay slow and manual.

03

No safe way to practice

Security teams rarely get to rehearse against a live, adapting adversary before facing one for real.

04

A widening skills gap

Traditional training — slides, static labs — doesn’t build the split-second judgment real incidents demand.

Proposed Solution

An adaptive simulator where two intelligent agents compete — while human users make the calls



Hacker Agent

Continuously probes the simulated network, choosing and evolving attack strategies based on what defenses it encounters.



Defender Agent

Detects, analyzes, and counters threats autonomously — learning from each exchange to respond faster next round.



Human Users

One or more users act as Security Manager, overseeing the battlefield — approving actions, allocating resources, and making calls the Defender can't make alone.

THE ROUND LOOP

1 **Hacker attacks** → 2 **Defender + Human respond** → 3 **Outcome scored** → 4 **Both sides adapt**

Impact

What changes once teams train against a live, adapting opponent



Faster Response Reflexes

Trains defenders to make rapid decisions under pressure, closing the gap between detection and action.



Realistic, Repeatable Training

Every session pits users against a freshly adapting attacker — no two runs play out the same way.



Benchmarking Defense Strategies

Generates data on which defensive tactics hold up against evolving AI-driven attacks.



Wider Security Awareness

Makes cybersecurity concepts tangible and engaging for students, not just specialists.

Project Budget

Indicative cost allocation for building and running the prototype (adjust to your team's actuals)



Conclusion

Hacker vs Defender reframes cybersecurity training as a live contest of adapting intelligence — turning "Are we secure?" into a rehearsable, measurable question.

- Makes real-time defense reflexes trainable, not theoretical
- Puts human judgment back at the center of every decision
- Creates a continuously evolving benchmark as both sides learn

"Can your defense move faster than the attacker can adapt?"

Let's find out.



THANK YOU

HACKER VS DEFENDER · QUESTIONS?